

Shubham Gupta

Jaipur, India
+91-7014524673

shubhamxgupta1@gmail.com — GitHub — LinkedIn — Portfolio — LeetCode

SUMMARY

AI/ML Engineer with demonstrated expertise in computer vision, deep learning, and real-time systems. Proven track record in designing end-to-end intelligent solutions—from model training and optimization to production deployment. Led cross-functional teams in research environments. Proficient in TensorFlow, OpenCV, and generative AI frameworks. Passionate about solving complex technical problems through data-driven design and scalable system engineering.

EDUCATION

Vellore Institute of Technology, Chennai B.Tech in Computer Science & Engineering (AI & ML) <i>CGPA: 8.5/10 (6 semesters completed)</i>	<i>Chennai, India</i> <i>2023–2027</i>
Maheshwari Public School, Jaipur Class XII (CBSE) – 80% Class X (CBSE) – 90%	<i>Jaipur, India</i> <i>2022–2023</i> <i>2020–2021</i>

EXPERIENCE

Walmart Global Tech <i>Software Engineer Intern – Sam’s Club</i>	<i>On-site</i> <i>May 2026 – Jul 2026</i>
- Built a secure internal merchandise website with Single Sign-On (SSO) authentication and role-based access control; collaborated with infrastructure teams to define monitoring metrics and optimize backend service reliability. - Architected a real-time monitoring dashboard for cross-microservice API call tracking and service-to-service communication patterns, providing operational visibility into system health and performance bottlenecks.	
Samsung PRISM <i>Team Lead – Research Intern</i>	<i>Remote</i> <i>Aug 2025 – May 2026</i>
- Led a 4-person team in building an automated XML-to-Jetpack Compose migration tool that reduced UI modernization effort by 40%; engineered Kotlin AST parsing to detect security vulnerabilities (hardcoded secrets, PII) pre-migration, achieving zero critical vulnerabilities in beta. - Designed a semantic mapping pipeline that generates idiomatic Compose components from complex nested XML layouts, enabling reliable conversion where standard open-source tools failed; conducted code reviews and mentored junior engineers on Android security best practices.	

PROJECTS

Real-Time AI Gym Coach & Form Analyzer — <i>Python, MediaPipe, OpenCV, Streamlit, Groq LLM, SQLite</i> Live Demo — GitHub	- Engineered end-to-end computer vision system using MediaPipe Pose (33 3D keypoints) and OpenCV for real-time exercise form analysis across 5+ exercise types; developed custom biomechanical algorithms achieving 95%+ form evaluation accuracy and real-time rep counting. - Optimized video processing pipeline with streamlit-webrtc and frame buffering to achieve sub-100ms latency on standard webcams without GPU acceleration; deployed containerized Streamlit Cloud application with 99%+ uptime and automated error recovery. - Integrated generative AI feedback loop using Groq LLM API and gTTS for personalized form corrections; built SQLite analytics layer for historical workout logging, progress tracking, and long-term performance visualization.
Samsung PRISM – XML-to-Jetpack Compose Migration Tool — <i>Kotlin, AST Parsing, Semantic Analysis</i> Download — GitHub	- Architected automated code transformation pipeline reducing Android UI modernization effort by 40%; designed Kotlin AST parsing engine to detect security vulnerabilities (hardcoded API keys, PII exposure) pre-migration, achieving zero critical vulnerabilities across 1000+ components. - Engineered semantic mapping layer capturing UI intent for intelligent component generation with proper theming, state management, and accessibility—enabling reliable conversion of complex nested XML layouts where standard open-source tools failed. - Maintained design-system-compliant idiomatic output; handled edge cases through collaborative validation and testing with product teams.

TECHNICAL SKILLS

AI/ML Frameworks: PyTorch (Advanced), TensorFlow (Intermediate), Scikit-Learn, Transformers, GANs Computer Vision & Signal Processing: OpenCV, MediaPipe Pose, Real-time pose estimation Generative AI & LLM: Prompt engineering, LLM integration, gTTS (Text-to-Speech) Data & Analytics: Pandas, NumPy, Data preprocessing, Feature engineering, Model evaluation Programming Languages: Java (Expert), Python (Advanced), Kotlin, c++, JavaScript, SQL Web & Backend: FastAPI (Advanced), Django, Flask, Node.js, Next.js, Streamlit, streamlit-webrtc Databases & Cloud: PostgreSQL, SQLite, MongoDB, Firebase, Google Cloud Platform, Docker Android / Mobile: Kotlin, Jetpack Compose, Static analysis, AST parsing, Android security Developer Tools: Git, Linux, Docker, Postman, VS Code, Jupyter Notebooks

ACHIEVEMENTS & RECOGNITION

World Rank 31 – CodeChef Weekly Contest 159 (3-Star Coder) — demonstrates advanced algorithmic problem-solving and competitive programming mastery. Top 10% on LeetCode – 500+ problems solved; strong grasp of data structures, algorithms, and system design. Google Cloud Arcade Champion 2024 – recognized expertise in cloud-native application development and infrastructure. Certification – IBM certified GenAI and Oracle certified Data Science expert.
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